

Subnetting Cheat Sheet

IPv4 CIDR, masks, and host counts at a glance

A CIDR prefix (/n) states how many leading bits of a 32-bit IPv4 address identify the network. The remaining bits identify hosts within that network.

Prefix	Subnet mask	Total addresses	Usable hosts
/8	255.0.0.0	16,777,216	16,777,214
/16	255.255.0.0	65,536	65,534
/20	255.255.240.0	4,096	4,094
/22	255.255.252.0	1,024	1,022
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2
/31	255.255.255.254	2	2 (RFC 3021 point-to-point)
/32	255.255.255.255	1	1 (host route)

How to calculate

1. Network address = IP address AND subnet mask.
2. Broadcast address = network address with all host bits set to 1.
3. Usable range = every address between network and broadcast, excluding both (except for /31 and /32, which are special cases).
4. Total addresses = 2 raised to the number of host bits (32 minus the prefix).

Private address ranges (RFC 1918)

Range	CIDR	Typical use
10.0.0.0 - 10.255.255.255	10.0.0.0/8	Large private networks
172.16.0.0 - 172.31.255.255	172.16.0.0/12	Medium private networks
192.168.0.0 - 192.168.255.255	192.168.0.0/16	Home and small office networks

